

Alvaro Perdomo Gutierrez

Software Engineer

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PROFESSIONAL SUMMARY

Full Stack Developer with **12+** years across eCommerce, healthcare, and gaming, building scalable enterprise applications in JavaScript, Python, and cloud infrastructure. Proficient in ServiceNow development, API design, system integrations, and microservices architecture. Experienced delivering customer-facing platforms and backend systems with strong focus on enterprise-grade reliability and maintainability.

SKILLS

Languages: JavaScript, TypeScript, Python, Java, Go, SQL, HTML, CSS

Backend: Node.js, Express, FastAPI, Django, Spring Boot, NestJS, GraphQL, RabbitMQ, Kafka

Frontend: React, Next.js, Angular, Vue.js, Redux, Tailwind CSS, Material-UI, webpack

ServiceNow: ServiceNow Platform, ServiceNow Workflows, Service Portal, ITSM, CMDB, Flow Designer, Business Rules, UI Builder

Database: PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch, DynamoDB, Firebase

Testing: Jest, Pytest, Mocha, Cypress, Selenium, JUnit, Vitest

Cloud & DevOps: AWS, Google Cloud Platform, Kubernetes, Docker, GitHub Actions, Jenkins, Terraform, CloudWatch, Datadog

Others: RESTful API, Microservices, System Integration, OAuth2, JWT, Git, GitHub, Jira, Agile, CI/CD, OpenTelemetry

EXPERIENCE

Leanware

May 2024–Present

Senior Software Engineer

- Led the design and development of cloud-based SaaS applications across multiple industry verticals including eCommerce, healthcare, and financial services, using **JavaScript**, **React**, and **Node.js** to build responsive, scalable front-end and back-end systems that served internal and external stakeholders.
- Spearheaded full-stack integration work connecting enterprise platforms through **RESTful APIs** and event-driven architectures, orchestrating data synchronization between disparate systems while maintaining data consistency and HIPAA compliance where required.
- Architected and deployed microservices infrastructure on **Google Cloud Platform** and **Kubernetes**, establishing automated **CI/CD** pipelines with **GitHub Actions** that reduced deployment friction and enabled the team to ship features twice as fast as previous deployment cadence.

- Engineered backend services using **Node.js** and **Express** to support AI-driven features across multiple products, integrating **PostgreSQL** and **Redis** for high-throughput data persistence and caching, handling 50K+ concurrent requests under peak load.
- Implemented comprehensive observability using **OpenTelemetry** and **Datadog**, which accelerated incident diagnosis and reduced mean-time-to-recovery from 2 hours to under 15 minutes across production systems.
- Collaborated with product and design teams to refactor legacy monolithic applications into modular components using **React** and **TypeScript**, reducing bundle size by **28%** and improving time-to-interactive for customer-facing dashboards.
- Mentored three junior developers on full-stack patterns, code review processes, and production debugging techniques, resulting in team members shipping independent features within 6 weeks.
- Established API contract testing using **Jest** and request validation schemas, eliminating integration bugs that had previously leaked into staging and production.
- Upgraded database connection pooling and query optimization across **PostgreSQL** instances, cutting average query latency by **35%** and reducing database CPU utilization during peak traffic windows.
- Shipped automated end-to-end test suites using **Cypress** covering critical user journeys, which caught a regression in the checkout flow before production rollout and prevented estimated revenue loss.
- Configured OAuth2 authentication and JWT token management across API services, ensuring secure inter-service communication and supporting OAuth2 flows for third-party integrations with financial partners.

ThredUp

Mar 2022–May 2024

Senior Software Engineer

- Drove the end-to-end integration of Lily AI visual recognition into the eCommerce platform, building **JavaScript** and **TypeScript** components in **React** that surfaced AI-powered product categorization and tagging to 275K images weekly, directly contributing to a **15%** lift in sell-through and **2.8%** iOS conversion improvement.
- Played a key role in migrating the ML feature platform from legacy infrastructure to **Databricks**, architecting data pipelines in **Python** that reduced model deployment time from 2 quarters to 1 month and accelerated MVP iteration cycles from 2 months to 2 weeks.
- Built full-stack shopping experience features including Image Search and Style Chat, developing **Node.js** backend services to orchestrate model inference calls, caching results in **Redis**, and exposing APIs consumed by **React** frontend components that shipped to production in August 2024.
- Optimized the proprietary ML visual recognition pipeline by refactoring **Python** preprocessing steps and tuning inference parameters, which improved model throughput from 80K to 120K daily product photos processed without increasing infrastructure costs.
- Engineered auto-pricing algorithm integrations that analyzed 172M+ inventory items across 55K brands, writing **SQL** queries against **PostgreSQL** data warehouse to surface pricing recommendations and influencing **12%** average price optimization.

- Implemented comprehensive error handling and retry logic for third-party ML API calls using **Express** middleware, adding exponential backoff and circuit breaker patterns that reduced downstream API failures by **40%** and improved system resilience.
- Designed and implemented feature flags using a custom **JavaScript** library built on **Redis**, enabling canary deployments and A/B testing for high-risk features without requiring full re-deployment.
- Collaborated with data science and product teams to establish ML evaluation metrics dashboards using **Grafana** and **Prometheus**, surfacing model performance degradation in real-time and enabling proactive model retraining decisions.
- Led refactoring of tightly coupled database queries into efficient **GraphQL** resolvers, reducing N+1 query patterns and cutting API response times by **30%** for the seller dashboard.
- Performed on-call rotation for production incidents, diagnosing and resolving system failures including a memory leak in image processing workers that was consuming 8GB daily, implementing structured logging with **OpenTelemetry** to catch similar issues before production impact.

Spring Health

Nov 2020–Mar 2022

Software Engineer

- Contributed to scaling a precision-matching ML platform supporting 2M+ healthcare members across 6x growth phase, building **Python** backend services using **FastAPI** that maintained clinical validity and compliance with healthcare regulations while tripling query throughput.
- Engineered the global Care Navigation platform using **TypeScript**, **React**, and **Node.js** to support dependent eligibility logic, localization for 12+ languages, and jurisdiction-aware matching rules that enabled Fortune 500 clients including PepsiCo and Instacart to manage distributed workforce benefits.
- Designed HIPAA-compliant infrastructure on **AWS** including encrypted **RDS** databases, **VPC** isolation, and audit logging that securely onboarded enterprise customers at significant scale, completing compliance audits 3 weeks ahead of schedule.
- Implemented operational ML features including dropout-risk prediction and demand forecasting models in **Python** using **scikit-learn** and **Pandas**, which improved care provider scheduling accuracy by **18%** and reduced member no-show rates.
- Built robust integration layers connecting disparate healthcare systems via **REST APIs**, implementing idempotency and retry patterns that ensured reliable data exchange and eliminated duplicate care matching across provider networks.
- Optimized database queries across millions of care records by refactoring **SQL** aggregations and adding appropriate indexes, reducing Care Navigation platform query latency from 2.5s to 600ms and improving user experience during peak load.
- Established observability infrastructure using **CloudWatch** and custom dashboards to track member matching latency and system health metrics across the platform, enabling oncall response to infrastructure anomalies within 5 minutes.
- Collaborated with clinical and product teams to validate that platform scaling maintained matching accuracy and member satisfaction scores, documenting data governance practices that set standards for future healthcare platform work.

Tencent Games

Software Engineer

Nov 2014–Nov 2020

- Built backend systems for Honor of Kings, a top mobile MOBA that scaled to 100M daily active users by November 2020, using **Java** and **C++** to implement game server logic, player matchmaking, and real-time synchronization handling millions of concurrent players.
- Developed core infrastructure components for PUBG Mobile using **C++** and Unreal Engine 4, including game state replication and network optimization that enabled 100-player battle royale gameplay with sub-100ms latency across global regions.
- Engineered the Tencent Gaming Buddy emulator using **C++** and **Java**, creating a lightweight Android runtime that improved developer productivity and reduced emulation overhead by **45%** compared to existing solutions.
- Implemented anti-cheat systems using kernel-level detection and behavioral analysis in **C++**, protecting competitive rankings and preventing account compromise — the solution reduced detected cheating incidents by **78%** across live titles.
- Designed and maintained high-throughput data storage layer using **TcaplusDB**, managing 40M+ daily active players' session state and game progression data with sub-10ms latency guarantees across distributed cluster infrastructure.
- Collaborated on cross-platform development for Arena of Valor, porting Honor of Kings gameplay to console and PC using network abstraction layers, enabling seamless matchmaking and progression sharing between mobile and desktop players.
- Optimized in-game communication using **GVoice** voice service, integrating low-latency audio codecs and echo cancellation that improved voice call stability during intense gameplay moments, reducing call drop rate from **8%** to **1.2%**.
- Contributed to the co-development of Contra Returns with Konami in 2017, implementing server-side weapon balancing and player progression systems that prepared the title for esports tournament play later that year.
- Established automated testing infrastructure using **Jenkins** and custom test harnesses to validate game balance changes before production deployment, catching exploitable bugs in economy systems before players could abuse them.

EDUCATION

The Hong Kong Polytechnic University
Bachelor of Science in Computer Science

Sep 2010–Nov 2014